

Sylow Theorems

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The structural classification of finite groups is a central theme in modern algebra.

While Lagrange's Theorem provides a necessary condition for the existence of subgroups stating that the order of a subgroup must divide the order of the group (it does not guarantee their existence). The converse of Lagrange's Theorem is false in the general case.

In this project, we explore the partial converse to Lagrange's Theorem provided by the Sylow Theorems. These theorems allow us to determine the existence of subgroups of prime-power order and provide information about their number and conjugation properties. To arrive at these powerful results, we first establish the necessary machinery of Group Actions and Conjugacy Classes. Finally, we demonstrate the utility of these theorems by analyzing the structure of a group of order 40.

1 Preliminaries

In this section, we provide all the theorems and the definitions that we have studied in Modern Algebra, and Discrete Mathematics courses that we used in the next sections. For the sake of time, we omit the proof of any previously proved theorem or corollary in the Lecture Notes.

Definition 1.1: Equivalence relation

A relation R on a set A is called an *equivalence relation* if it satisfies the following three properties:

- For any $a \in A$, we have that aRa (Reflexivity)
- For any $a, x \in A$, we have that if aRx , then xRa (Symmetry)
- For any $a, x, y \in A$, we have that if aRx and xRy , then aRy (Transitivity)

Definition 1.2: Equivalence class

Let R be an equivalence relation on set A , and let $a \in A$. The equivalence class of a is the set of all elements in A that are equivalent to a . That is, denoted by $[a]_R$ or $[a]$.

$$[a] = \{b \in A \mid b \sim a\}$$

Theorem 1.1

Suppose R is an equivalence relation on a set A . Then we can express A as

$$A = [x_1] \cup [x_2] \cup [x_3] \cup \cdots \cup [x_n]$$

where $x_1, x_2, x_3, \dots, x_n \in A$ and the equivalence classes $A = [x_1] \cup [x_2] \cup [x_3] \cup \cdots \cup [x_n]$ are pairwise disjoint. In other words, A is the union of the distinct equivalence classes of R

Definition 1.3: Center of Group

The center of a group G is the set

$$Z(G) = \{g \in G \mid \forall h \in G, gh = hg\}.$$

Definition 1.4: Centralizer

The Centralizer of an element $g \in G$ is the set

$$C(g) = \{a \in G \mid ag = ga\}.$$

Theorem 1.2: Lagrange's Theorem

The order of a subgroup of a finite group divides the order of the group.

Definition 1.5: Normalizer

Let H be a subgroup of a group G . The normalizer of H , denoted by $N(H)$, is the set of all elements in G that preserve the set H under conjugation.

$$N(H) = \{g \in G \mid gHg^{-1} = H\}.$$

Definition 1.6: Normal subgroup

A subgroup N of a group G is called **normal** if for any $g \in G$ and any $n \in N$ we have that $gn g^{-1} \in N$. We denote this by $N \trianglelefteq G$ or $N \triangleleft G$.

Theorem 1.3

Let $Z(G)$ be the center of a group G . if $G/Z(G)$ is cyclic, then G is abelian.

Theorem 1.4: Cauchy's Theorem

Let G be a finite abelian group, and suppose that p is a prime that divides $|G|$. Then G has an element of order p .

Definition 1.7: p-Group

Let p be a prime number and let G be a finite group. We call G a *p-Group* if $|G| = p^n$, where $n \geq 1$.

Theorem 1.5

Let $\phi : G \rightarrow H$ be a homomorphism, and let L be a subgroup of H . Then the subset $\phi^{-1}(L) = \{g \in G \mid \phi(g) \in L\}$ is a subgroup of G .

Theorem 1.6

Let N be a normal subgroup of group G . Every subgroup of G/N has the form H/N , where H is a subgroup of G .

Proof.

Let $\bar{K}$ be a subgroup of the quotient group G/N . We define the set H to be the preimage of $\bar{K}$ under a homomorphism $\pi : G \rightarrow G/N$ such that $\pi(g) = gN$. That is:

$$H = \{g \in G \mid gN \in \bar{K}\}.$$

Since $\bar{K}$ is a subgroup, it contains the identity element of G/N , which is $N = eN$. Thus, $e \in H$. Let $a, b \in H$, then $aN \in \bar{K}$ and $bN \in \bar{K}$. Since $\bar{K}$ is a group, $(aN)(bN) = (ab)N \in \bar{K}$. Thus, $ab \in H$. Let $a \in H$, then $aN \in \bar{K}$. Since $\bar{K}$ is a group, $(aN)^{-1} = a^{-1}N \in \bar{K}$. Thus $a^{-1} \in H$. Hence, H is a subgroup of G .

For any $n \in N$, we have $nN = N$. Since $\bar{K}$ is a subgroup, it contains the identity coset. Therefore, $nN \in \bar{K}$ for all $n \in N$, which implies $N \subseteq H$. By the definition of H , the set of cosets $H/N = \{hN \mid h \in H\}$ is exactly the set of elements in $\bar{K}$. Thus, $\bar{K}$ corresponds to the subgroup H of G containing N . ■

2 Group Actions

With the fundamental definitions of subgroups and quotient groups established, we now turn to the tool that drives the proofs of the Sylow Theorems, Group Actions.

Group actions allow us to translate abstract group properties into concrete combinatorial data by examining how a group permutes the elements of a set.

Definition 2.1: Group Actions

Let G be a group and X be a non-empty set. A *group action* of G on X is a map $G \times X \rightarrow X$, denoted by $(g, x) \mapsto g \cdot x$, satisfying:

- $e \cdot x = x$ for all $x \in X$, where e is the identity of G .
- $(gh) \cdot x = g \cdot (h \cdot x)$ for all $g, h \in G$ and $x \in X$.

Examples of Group Actions

- **Matrix-Vector Multiplication:** Let $G = GL_n(\mathbb{R})$ be the general linear group and $X = \mathbb{R}^n$. The group acts on the vector space by matrix multiplication defined by $A \cdot v = Av$ for all $A \in G$ and $v \in \mathbb{R}^n$.
- **Symmetries of a Square:** Let $G = D_4$ and $X = \{1, 2, 3, 4\}$ be the vertices of a square. The group acts on the set by permuting the vertices through rotations and reflections.
- **Conjugation:** Let G be any group and $X = G$. The action is defined by conjugation: $g \cdot x = gxg^{-1}$.

Lemma 2.1: Orbits

Let G be a group that acts on a set X . We say that $x, y \in X$ are **G -equivalent** if there is $g \in G$ such that $y = g \cdot x$. G -equivalence is an equivalence relation and the classes are called **orbits**. We denote it by $\mathcal{O}_x$ for the orbit containing x .

Proof.

Let $\sim$ be the G -equivalence relation on a set X . We show that it is:

- Reflexive: For any $x \in X$, $x = e \cdot x$, so $x \sim x$.
- Symmetric: If $x \sim y$, then $y = g \cdot x$. Multiplying both sides by g^{-1} gives $x = g^{-1} \cdot y$, so $y \sim x$.
- Transitive: If $x \sim y$ ($y = g \cdot x$) and $y \sim z$ ($z = h \cdot y$), then $z = h \cdot (g \cdot x) = (hg) \cdot x$. Since $hg \in G$, $x \sim z$.

Thus, $\sim$ is an equivalence relation. Then, its equivalence classes are precisely the orbits $\mathcal{O}_x$. By Theorem 1.1, these classes partition the set X . Thus, the total size of X is the sum of the sizes of these disjoint orbits. ■

Definition 2.2: Stabilizer

Let a group G act on a set X . The *stabilizer* of an element $x \in X$ is the set of elements in G that fix x :

$$G_x = \{g \in G \mid g \cdot x = x\}.$$

Example: Consider the action of D_4 on the vertices of a square $X = \{1, 2, 3, 4\}$.

- Let $x = 1$ (the top-left vertex).
- The identity e fixes 1 ($e \cdot 1 = 1$).
- The reflection D across the main diagonal fixes 1 ($D \cdot 1 = 1$).
- No other element fixes 1. For instance, a 90° rotation moves 1 to 2.

Thus, the stabilizer of 1 is the subgroup $G_1 = \{e, D\}$.

Lemma 2.2

Let G be a group acting on a set X . For any element $x \in X$, the stabilizer $G_x = \{g \in G \mid g \cdot x = x\}$ is a subgroup of G .

Proof.

We check the three conditions for a subgroup:

1. **Identity:** By the definition of a group action, $e \cdot x = x$ (where e is the identity of G). Thus, $e \in G_x$, so the stabilizer is non-empty.
2. **Closure under the operation:** Let $g, h \in G_x$. By definition, $g \cdot x = x$ and

$h \cdot x = x$. We check the product gh :

$$(gh) \cdot x = g \cdot (h \cdot x) = g \cdot (x) = x.$$

Thus, $gh \in G_x$.

3. **Inverses:** Let $g \in G_x$, so $g \cdot x = x$. We consider the action of g^{-1} on both sides:

$$g^{-1} \cdot (g \cdot x) = g^{-1} \cdot x.$$

Using the compatibility property on the left side:

$$(g^{-1}g) \cdot x = g^{-1} \cdot x \implies e \cdot x = g^{-1} \cdot x \implies x = g^{-1} \cdot x.$$

Thus, $g^{-1} \in G_x$.

Therefore, G_x is a subgroup of G . ■

Theorem 2.1: The Orbit-Stabilizer Theorem

Let G be a finite group acting on a finite set X . For any $x \in X$, the size of the orbit of x is equal to the index of its stabilizer in G :

$$|\mathcal{O}_x| = |G : G_x|.$$

Consequently, $|G| = |\mathcal{O}_x| \cdot |G_x|$.

Proof.

Let S be the set of left cosets of the stabilizer G_x in G . We define a map $\psi : \mathcal{O}_x \rightarrow S$ by

$$\psi(g \cdot x) = gG_x.$$

We must show that this map is well-defined and is a bijection. First, we show that ψ is well-defined, that is, it does not depend on the choice of a coset representative. Suppose $g \cdot x = h \cdot x$ for some $g, h \in G$. Observe that

$$g \cdot x = h \cdot x \implies h^{-1} \cdot (g \cdot x) = x \implies (h^{-1}g) \cdot x = x.$$

Thus, $h^{-1}g \in G_x$, which implies $gG_x = hG_x$.

Next, we show that ψ is injective. Suppose $\psi(g \cdot x) = \psi(h \cdot x)$. See that

$$gG_x = hG_x \implies h^{-1}g \in G_x \implies (h^{-1}g) \cdot x = x.$$

Acting by h on both sides gives $g \cdot x = h \cdot x$. Thus, the map is injective.

Finally, we show that ψ is surjective. Let gG_x be any left coset in S . By definition of the orbit, $y = g \cdot x$ is in $\mathcal{O}_x$, and $\psi(y) = gG_x$. Thus, the map is surjective. Since ψ is a bijection, the number of elements in the orbit $|\mathcal{O}_x|$ equals the number of left cosets $|S|$. By Lagrange's Theorem, the number of cosets is the index $|G : G_x|$. Therefore, we proved that $|\mathcal{O}_x| = |G : G_x|$. ■

Definition 2.3: Fixed Points

Let a group G act on a set X . An element $x \in X$ is called a *fixed point* if $g \cdot x = x$ for all $g \in G$, that is, $G_x = G$. The set of all fixed points is denoted by $\text{Fix}(G)$.

Lemma 2.3

Let P be a finite p -group acting on a finite set X . Then, the number of fixed points $|\text{Fix}(P)|$ is congruent to the total size of the set modulo p :

$$|\text{Fix}(P)| \equiv |X| \pmod{p}.$$

Proof.

By Lemma 2.1, the orbits of the action partition the set X . Thus, we can write the total size of X as the sum of the sizes of these disjoint orbits:

$$|X| = \sum |\mathcal{O}_x|.$$

By the Orbit-Stabilizer Theorem, the size of any orbit $|\mathcal{O}_x|$ must divide the order of the group $|P|$. Since P is a p -group, $|P| = p^n$, every orbit size must be a power of p . We distinguish two cases for the orbits:

- **Case 1** ($|\mathcal{O}_x| = 1$): An orbit has size 1 if and only if $g \cdot x = x$ for all $g \in P$. These are exactly the elements in $\text{Fix}(P)$.
- **Case 2** ($|\mathcal{O}_x| > 1$): If an orbit has size greater than 1, it must be a multiple of p . Thus, $|\mathcal{O}_x| \equiv 0 \pmod{p}$.

Therefore, when we take the sum modulo p , all orbits of size greater than 1 vanish, leaving only the sum of the fixed points:

$$|X| = \sum_{x \in \text{Fix}(P)} 1 + \sum_{|\mathcal{O}_x| > 1} |\mathcal{O}_x| \equiv |\text{Fix}(P)| + 0 \pmod{p}.$$

$$|X| \equiv |\text{Fix}(P)| \pmod{p}.$$
■

Lemma 2.4

Let P be a Sylow p -subgroup of a finite group G . Then the index of P in G is not divisible by p :

$$p \nmid |G : P|.$$

Proof.

Let $|G| = p^n m$ where $p \nmid m$. By definition of a Sylow p -subgroup, $|P| = p^n$. By Lagrange's Theorem:

$$|G : P| = \frac{|G|}{|P|} = \frac{p^n m}{p^n} = m.$$

Since p does not divide m , p does not divide the index $|G : P|$. ■

3 Conjugacy Classes

Having established the general theory of group actions, we now focus on a specific and highly significant action, the action of a group on itself by conjugation. This action partitions the group into disjoint conjugacy classes. The analysis of these classes, encapsulated in the "Class Equation," provides a powerful arithmetic relationship between the order of a group and the size of its center, which is instrumental in proving the first Sylow Theorem.

Definition 3.1: Conjugacy class

Let a, b be elements of group G . We say that a, b are *conjugate* in G and we say that b is the *conjugate* of a if and only if $gag^{-1} = b$ for some $g \in G$. The set of all conjugates of an element $a \in G$, denoted by $\text{cl}(a) = \{gag^{-1} \mid g \in G\}$, is the **Conjugacy class of a** .

Lemma 3.1

Conjugacy is an *equivalence relation* on a group.

Proof.

Let R be the conjugacy relation and let G be a group. We show that it is:

- Reflexive: Let $a \in G$. See that $ea e^{-1} = a$, so, every element is conjugate to itself and the relation is reflexive.
- Symmetric: Let $a, b \in G$. Suppose that b is conjugate of a , that is, there exists $g \in G$ such that $gag^{-1} = b$. Conjugate b by g^{-1} , $g^{-1}bg = g^{-1}gag^{-1}g = a$. Thus, a is conjugate to b and R is symmetric.

- **Transitive:** Let $a, b, c \in G$. Suppose that a is conjugate to b , then there exists a $g \in G$ such that $a = bgg^{-1}$, b is conjugate to c , then there exists an $h \in G$ such that $b = hch^{-1}$. Observe that, $a = bgg^{-1} = ghch^{-1}g^{-1} = (gh)c(gh)^{-1}$, and a is conjugate to c . Thus, R is transitive. ■

See that, by Theorem 1.1, we can partition groups into disjoint conjugacy classes.

- **Example**

In D_4 , we have:

$$\begin{aligned} \text{cl}(R_{90}) &= \{R_{90}R_{90}R_{90}^{-1}, R_{180}R_{90}R_{180}^{-1}, R_{270}R_{90}R_{270}^{-1}, R_0R_{90}R_0^{-1}, \\ &HR_{90}H^{-1}, VR_{90}V^{-1}, DR_{90}D^{-1}, LR_{90}L^{-1}\} = \{R_{90}, R_{270}\} \end{aligned}$$

Lemma 3.2

$\forall x \in Z(G)$ we have that $\text{cl}(x) = \{x\}$.

Proof.

Suppose we have a group G , pick an element $a \in Z(G)$. By definition of the center of the group, a commutes with any element in G . Conjugate a by $g \in G$, that is, gag^{-1} . Since a commutes with any element in the group, observe

$$gag^{-1} = agg^{-1} = ae = a.$$

Thus, the conjugacy class of an element in the center of the group is the singleton that contains this element. ■

The next theorem shows a relationship between the conjugacy class of an element a and the index of the centralizer of a .

Theorem 3.1

Let G be a finite group and let a be an element of G . Then, $|\text{cl}(a)| = |G : C(a)|$.

Proof.

Suppose we have a finite group G and let a be an element of G . To prove that the order of the set of conjugates of an element equals the index of that element centralizer in G we need to find a bijection between the set of conjugates and the set of cosets of the centralizer, let S be the set of left cosets of $C(a)$ in G . Suppose we have the function $\phi : S \rightarrow \text{cl}(a)$ such that $\phi(xC(a)) = xax^{-1}, x \in G$.

First, we need to show that ϕ is well defined, that is, if $xC(a) = yC(a)$, then $\phi(xC(a)) = \phi(yC(a))$. Let x, y be elements in the group G and suppose that $xC(a) =$

$yC(a)$. So, $y^{-1}x \in C(a)$ and $x = y(y^{-1}x)$. Hence, since $y^{-1}x$ commutes with a we get that:

$$\phi(xC(a)) = \phi(y(y^{-1}x)C(a)) = y(y^{-1}x)a(y^{-1}x)^{-1}y^{-1} = yay^{-1}.$$

Since we get that $\phi(xC(a)) = \phi(yC(a))$, then the function is well defined.

Next, we show ϕ is injective. Suppose $\phi(xC(a)) = \phi(yC(a))$, then, $axa^{-1} = yay^{-1}$. Multiply both sides by y^{-1} from the left and by x from the right. Observe that $y^{-1}xa = ay^{-1}x$, so $y^{-1}x$ commutes with a and $y^{-1}x \in C(a)$. Since $y^{-1}x \in C(a)$ then, $xC(a) = yC(a)$ as required. Thus, the function ϕ is injective.

Finally, we show that ϕ is surjective. Let b be any element in the conjugacy class $\text{cl}(a)$. By definition, $b = gag^{-1}$, $g \in G$. We can see the coset $gC(a)$ maps to b . Clearly, the function is surjective.

Therefore, we proved that the function $\phi : S \rightarrow \text{cl}(a)$ such that $\phi(xC(a)) = axa^{-1}$ is bijective and the order of the conjugacy class of an element a is the index of the centralizer of that element. ■

Corollary 3.1

Let G be a finite group and let a be an element in G . $|\text{cl}(a)|$ divides $|G|$, $a \in G$.

Proof.

By the previous theorem, we know that $|\text{cl}(a)| = |G : C(a)|$. We know by Lagrange's Theorem that the index of a subgroup divides the order of the group and we know that the centralizer of an element in the group is a subgroup. Hence, the order of the conjugacy class of a divides the order of the group G . ■

Corollary 3.2: The Class Equation

Let G be a finite group.

$$|G| = \sum |G : C(a)|,$$

where the sum runs over one element a from each conjugacy class of G .

Proof.

We know by Lemma 3.1 that the conjugacy classes partition the group, that is,

$$|G| = |\text{cl}(x_1)| + |\text{cl}(x_2)| + \cdots + |\text{cl}(x_n)|,$$

where $x_1, x_2, \dots, x_n$ are the representatives of the distinct conjugacy classes. By Theorem 3.1, the index of the centralizer of an element is the order of the conjugacy class of the same element. Thus, we can rewrite the class equation as $|G| = \sum |\text{cl}(a)|$. Since

the conjugacy class of an element in the center is the singleton that contains the same element, the order of this class is 1 and the sum of the sizes of these specific classes is exactly the order of the center of the group. Now, we can write the equation

$$|G| = |Z(G)| + \sum_{a \notin Z(G)} |G : C(a)|.$$

Example: Let $G = S_3$, which has order 6. The conjugacy classes are:

- $\{e\}$ (size 1)
- $\{(12), (13), (23)\}$ (size 3)
- $\{(123), (132)\}$ (size 2)

The class equation states $|G| = \sum |cl(a)|$. Indeed, we see that $6 = 1 + 3 + 2$.

Theorem 3.2

Let G be a p -Group, then $Z(G)$ has more than one element.

Proof.

Suppose the order of a p -Group G is p^n for some $n \geq 1$. We start by collecting the elements in the center, and write the class equation in the form

$$|G| = |Z(G)| + \sum_{a \notin Z(G)} |G : C(a)|$$

where the sum runs over all representatives of all conjugacy classes with more than one element. We know that $|G : C(a)| = |G|/|C(a)|$, and $C(a)$ is a subgroup of G . So each term in $\sum |G : C(a)|$ is of the form p^k , $k \geq 1$. Now, observe

$$|Z(G)| = |G| - \sum |G : C(a)|.$$

Since $|G|, \sum |G : C(a)|$ are divisible by p , then $|Z(G)|$ is divisible by p . Since $e \in Z(G)$, $Z(G)$ cannot be empty. Since $|Z(G)|$ is divisible by p and is non-zero, it must be at least p . Thus, $Z(G)$ has more than one element. ■

Corollary 3.3

if $|G| = p^2$, where p is a prime, then G is **abelian**.

Proof.

By Theorem 3.2 and Lagrange's Theorem, $|Z(G)|$ is either p or p^2 . If $|Z(G)| = p^2$, $G = Z(G)$ and we are done. Otherwise, suppose $|Z(G)| = p$. Then $|G/Z(G)| = p$ and $G/Z(G)$ is cyclic. By Theorem 1.3, G is abelian. ■

4 The Sylow Theorems

Remember that the inverse of Lagrange's Theorem is not valid. Suppose we have a group G of order n , if m divides n , this does not necessarily imply that G has a subgroup of order m . However, the Norwegian mathematician Ludwig Sylow (1832-1918) proved the next theorem, which is a partial inverse of Lagrange's Theorem.

Theorem 4.1: Sylow's First Theorem (1872)

Let G be a finite group and let p be a prime. If p^k divides $|G|$, then G has at least one subgroup of order p^k .

Proof.

We proceed by induction,

Base case: Suppose we have a group G and $|G|$ is divisible by $p^0 = 1$. It trivially follows that G has one subgroup of order 1, which is the identity.

Inductive step: Assume that the sentence is true for all group of order less than $|G|$. Then, we have two cases to investigate:

- **Case 1:** If G has a proper subgroup H such that $p^k \mid |H|$, then, by the inductive hypothesis, H has a subgroup of order p^k . This subgroup of order p^k is also a subgroup of G since H is a subgroup of G .
- **Case 2:** Otherwise, assume G does not have a proper subgroup with order divisible by p^k . Observe the class equation:

$$|G| = |Z(G)| + \sum |G : C(a)|,$$

where we run the sum over a representative of each conjugacy class $cl(a)$, and $a \notin Z(G)$. See that $|G| = |G : C(a)| \cdot |C(a)|$. We know the centralizer is a proper subgroup, so $p^k \nmid |C(a)|$, but $p \mid |G|$, so $p \mid |G : C(a)|$. From the class equation, $p \mid |Z(G)|$. By Cauchy's Theorem, there exist an element $x \in Z(G)$ such that it has order p . Since $x \in Z(G)$, it constructs a cyclic group $\langle x \rangle$ which is normal in G . We know that $|G|$ is divisible by p^k , that is, we can write $|G| = mp^k$. Now, we observe that the order of the quotient group $|G/\langle x \rangle| = mp^k/p = mp^{k-1}$, so it is divisible by p^{k-1} . By our inductive hypothesis, $G/\langle x \rangle$ has a subgroup of order p^{k-1} and by Theorem 1.6, we know that this subgroup has the form $H/\langle x \rangle$, where

H is a subgroup of G . Observe that the order of $H/\langle x \rangle = p^{k-1}$ and $|\langle x \rangle| = p$, so $|H| = p^k$. We found a subgroup of order p^k , which is a contradiction to our assumption. Therefore, H is the desired subgroup of order p^k and it exists in both cases if p^k divides $|G|$. ■

By Sylow's First Theorem, if we have a group G such that $|G| = 2^2 \cdot 5^3 \cdot 7^4$, then, G has at least one subgroup of each of the following orders: 2, 4, 5, 25, 125, 7, 49, 343, 2401. Certain subgroups that are guaranteed by Sylow's First Theorem are special in the theory of finite groups. Hence, they are given a special name, *Sylow p -subgroup*.

Definition 4.1: Sylow p -Subgroup

Let G be a finite group and let p be a prime. if p^k divides $|G|$ and p^{k+1} does not divide $|G|$, then any subgroup of G of order p^k is called a *Sylow p -Subgroup* of G .

Recall our previous example of G that has order $2^2 \cdot 5^3 \cdot 7^4$. By *Sylow p -subgroup* definition, we call any subgroup of order 8 a *Sylow 2-subgroup* of G , any subgroup of order 125 a *Sylow 5-subgroup* of G , and so on. Now, we can generalize Cauchy Theorem to include all finite groups.

Corollary 4.1: Cauchy's Theorem

Let G be a finite group, and suppose that p is a prime divisor of $|G|$. Then G has an element of order p .

Proof.

Suppose we have a finite group G such that $|G|$ is divisible by a prime p . By Sylow's First Theorem, we know that G has a subgroup H of order p . Since $|H| = p$ and p is a prime number, then H is cyclic and generated by a non-identity element. Let h be a generator of H ; the order of h is p . Thus, we found an element in G of order p . ■

Before moving to the two more Sylow Theorems, we introduce new terms.

Definition 4.2: Conjugate subgroups

Let H, K be subgroup of a group G . We say that H and K are conjugate in G if there is an element $g \in G$ such that $H = gKg^{-1}$.

Theorem 4.2: Sylow's Second Theorem

Let G be a finite group and let H be a subgroup of G . If $|H|$ is a power of a prime p , then H is contained in some Sylow p -Subgroup of G .

Proof.

By Sylow's First Theorem, Let P be a Sylow p -subgroup of G . Let X be the set of left cosets of P in G , given by $X = \{gP \mid g \in G\}$. We consider the action of H on X by left multiplication. By Lemma 2.3, the number of fixed points satisfies:

$$|\text{Fix}(H)| \equiv |X| \pmod{p}.$$

The size of the set is the index $|X| = |G : P|$. By Lemma 2.4, we know that p does not divide $|G : P|$. Therefore:

$$|X| \not\equiv 0 \pmod{p}.$$

This implies that $|\text{Fix}(H)| \not\equiv 0 \pmod{p}$, so the set of fixed points $\text{Fix}(H)$ cannot be empty.

Thus, there exists at least one coset $gP \in X$ that is fixed by H . That is, for all $h \in H$:

$$h \cdot (gP) = gP.$$

Multiplying by g^{-1} on the left implies:

$$g^{-1}hgP = P \implies g^{-1}hg \in P.$$

Since this holds for all $h \in H$, we conclude that $g^{-1}Hg \subseteq P$. Conjugating by g yields:

$$H \subseteq gPg^{-1}.$$

Since P is a Sylow p -subgroup, its conjugate gPg^{-1} has the same order as P and is therefore also a Sylow p -subgroup. Therefore, we proved that H is contained in the Sylow p -subgroup gPg^{-1} . ■

Theorem 4.3: Sylow's Third Theorem

Let p be a prime and let G be a group of order $p^k m$, where $p \nmid m$. Then the number n of Sylow p -Subgroups of G is equal to $1 \pmod{p}$ and divides m .

Proof.

Let S be the set of all Sylow p -subgroups of G , so $|S| = n$. By Sylow's Second Theorem, any two Sylow p -subgroups are conjugate, which implies that G acts transitively on S by conjugation. First, let $P \in S$. Since the action is transitive, the orbit of P is the entire set S . By the Orbit-Stabilizer Theorem:

$$|S| = |G : N(P)|,$$

where $N(P)$ is the normalizer of P in G . Since $P \leq N(P) \leq G$, we can use the multiplicative property of indices:

$$|G : P| = |G : N(P)| \cdot |N(P) : P|.$$

We know that $|G : P| = m$, since $|G| = p^k m$ and $|P| = p^k$. We substitute $n = |G : N(P)|$

$$m = n \cdot |N(P) : P|.$$

Since $|N(P) : P|$ is an integer, it follows that n divides m . Next, let the subgroup P act on the set S by conjugation. By Lemma 2.3:

$$|S| \equiv |\text{Fix}(P)| \pmod{p}.$$

Here, $\text{Fix}(P)$ is the set of Sylow p -subgroups Q that are normalized by P . Clearly, P normalizes itself, so $P \in \text{Fix}(P)$. Suppose there is another $Q \in \text{Fix}(P)$. Then $P \leq N(Q)$. Since Q is normal in $N(Q)$ and P is a Sylow p -subgroup of $N(Q)$, P must be contained in a conjugate of Q inside $N(Q)$. But since Q is normal, its only conjugate is Q itself. Thus $P \leq Q$. Since $|P| = |Q| = p^k$, we must have $P = Q$. Therefore, the only fixed point is P itself, so $|\text{Fix}(P)| = 1$. Substituting this back into the congruence:

$$n \equiv 1 \pmod{p}.$$

Corollary 4.2

A Sylow p -subgroup of a finite group G is a normal subgroup of G if and only if it is the only Sylow p -subgroup of G .

Proof.

For the backward direction, suppose P is the unique Sylow p -subgroup of G . Let $g \in G$. The conjugate subgroup gPg^{-1} has the same order as P ($|gPg^{-1}| = |P|$). By the definition of a Sylow p -subgroup, gPg^{-1} is also a Sylow p -subgroup. Since P is the *only* Sylow p -subgroup, we must have $gPg^{-1} = P$. Since this holds for all $g \in G$,

P is normal in G .

For the forward direction, suppose P is a normal Sylow p -subgroup of G . Let Q be any other Sylow p -subgroup of G . By Sylow's Second Theorem, P and Q are conjugate, meaning there exists $g \in G$ such that $Q = gPg^{-1}$. Because P is normal, we know that $gPg^{-1} = P$ for all $g \in G$. Therefore, $Q = P$. This implies that P is the unique Sylow p -subgroup of G . ■

5 An application of Sylow Theorems (analysis of a group of order 40)

To illustrate the power of the theory developed thus far, we conclude with a concrete application. We will use the Sylow Theorems to determine the structural properties of a group of a specific order, showing how numerical constraints can force the existence of normal subgroups.

Example

Let G be a group of order 40. By prime decomposition, $|G| = 40 = 2^3 * 5$. The first Sylow theorem tells us that this group has at least one subgroup of each of the following orders: 2, 4, 8, 5. We want to find the number of Sylow 5-subgroups. Using the third theorem, we know that the number of such subgroups, call it t , must divide 8 and $t \equiv 1 \pmod{5}$. Hence, the only available value of t is 1, so G has exactly one Sylow 5-subgroup. By Corollary 4.2, this subgroup is normal. Now, examine the number of Sylow 2-subgroups, call it s . We know that s divides 5, so it is either 1 or 5. Observe that both options satisfy $s \equiv 1 \pmod{2}$, and we cannot determine the exact number of Sylow 2-subgroups in G . If there is only one Sylow 2-subgroup then it is normal, otherwise, none of them is normal and all five can be obtained by conjugating any one of the five subgroups with elements of the group G . Finally, let K be the Sylow 5-subgroup and let H be any Sylow 2-subgroup. Since the $\gcd(|K|, |H|) = 1$, then their intersection must be the trivial group. Since K is normal, HK is a subgroup.